LOKESH NALLA

Full-Stack Developer | Machine Learning Engineer | AI Systems Builder

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Developer and Machine Learning Engineer with hands-on experience building scalable web applications, deep learning models, and AI-powered multi-agent systems. Proficient in Python, Django, JavaScript, and PyTorch. Demonstrated ability to deliver end-to-end software solutions from system design to deployment. Currently building InfraPulse, an autonomous multi-agent AI system for municipal infrastructure management using Groq API and Llama 4.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, SQL, C, C#

Web Technologies: HTML5, CSS3, Django, Node.js, React Native, Tailwind CSS, Bootstrap

AI / Machine Learning: PyTorch, ResNet50, VGG16, XGBoost, Random Forest, SVM, Transfer Learning, Groq API, Llama 4, Multi-Agent Systems, Prompt Engineering

Databases & Cloud: Supabase, SQL

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, PyCharm, Figma, Unity

Parallel Computing: OpenMP, Pthreads

Other: REST APIs, UI/UX Design, Game Development (Unity/C#)

WORK EXPERIENCE

Software Engineer Intern | Yugayatra *2024 | 2 Months*

- Launched Tutor Connect, a full-stack web platform connecting students with tutors, streamlining the student-tutor matching process through an intuitive digital interface.
- Engineered role-based access control, real-time scheduling logic, and dynamic user navigation flows using HTML, CSS, JavaScript, Tailwind CSS, and Supabase, cutting manual coordination effort by over 50%.
- Delivered a production-ready platform across the full development lifecycle — from requirements gathering and UI/UX wireframing through to deployment — within a 2-month internship period.

PROJECTS

InfraPulse — Autonomous Municipal Road Repair AI System | *2025 | Ongoing*

Python, Groq API, Llama 4 Scout, Llama 3, Multi-Agent Pipeline | github.com/loke2006/InfraPulse-ET-Hackathon

- Architected a 4-agent AI pipeline (VisionAgent, CostAgent, ComplianceAgent, OrchestratorAgent) for autonomous municipal tender generation; qualified for ET GenAI Hackathon 2026 Phase 2 out of hundreds of submissions.
- Integrated human-in-the-loop approval workflows, a full audit trail, and automated PDF work order generation to satisfy real-world municipal compliance requirements.
- Shipped an Engineer Control Room dashboard enabling real-time monitoring and control of all 4 agents, reducing manual oversight effort across the pipeline.

Full-Stack E-Commerce Platform | *2024*

Python, Django, JavaScript, Tailwind CSS, Supabase | github.com/loke2006

- Delivered a feature-complete e-commerce application supporting user authentication, product browsing, cart management, order processing, payment integration, and AI chatbot support.
- Designed a secure Django backend with a database-driven product management system and cart-to-order conversion pipeline, handling the complete purchase workflow end-to-end.
- Built a fully responsive frontend with Tailwind CSS, ensuring consistent cross-device compatibility across mobile and desktop.

Brain Tumor Classification — Deep Learning Model | *2024*

Python, PyTorch, ResNet50, VGG16, Transfer Learning | github.com/loke2006

- Trained a deep learning classifier to detect and categorize brain tumors from MRI scans using transfer learning on ResNet50 and VGG16, achieving strong classification accuracy on a multi-class medical dataset.
- Authored a skull stripping preprocessing pipeline that removes non-brain tissue from MRI input, improving model reliability even when eye regions are present in the scan.
- Boosted model performance through targeted hyperparameter tuning and data augmentation, reducing misclassification on edge-case MRI inputs.

Predictive Sentiment Analysis Model | *2024*

Python, XGBoost, Random Forest, SVM | github.com/loke2006

- Engineered a comparative analysis system across XGBoost, Random Forest, and SVM to classify customer review sentiment, achieving 95% accuracy on the test set.
- Conducted feature importance analysis and hyperparameter tuning across all 3 algorithms to isolate the key factors driving positive and negative customer sentiment.
- Produced an interpretable output layer that allows stakeholders to trace each sentiment prediction back to its specific causal features.

2D/3D Game Development Suite | *2023–2024*

Unity, C# | github.com/loke2006

- Shipped 3+ fully playable 2D games and an ongoing 3D adventure title in Unity, scripting complex game behaviors, physics interactions, and AI-driven NPC mechanics in C#.

EDUCATION

B.Tech in Computer Science and Engineering *2023 – 2027*

Amrita Vishwa Vidyapeetham, Amritapuri, Kerala | GPA: 6.59/10.0

Class 12 (Intermediate) *Completed 2022*

Resonance College, Telangana | Score: 87%

Class 10 (SSC) *Completed 2020*

Triveni School, Telangana | Score: 100%

CERTIFICATIONS & TRAINING

- Preprocessing for Machine Learning — DataCamp
- Machine Learning for Everyone — IBM
- Deep Learning in Python — DataCamp
- Understanding Prompt Engineering — DataCamp
- UI/UX Development — Teachnook
- Technology Job Simulation — Deloitte
- What is Generative AI? — LinkedIn Learning

ACHIEVEMENTS & ACTIVITIES

- Qualified for ET GenAI Hackathon 2026 Phase 2 with InfraPulse, an autonomous multi-agent AI system for municipal infrastructure management.
- Participated in L'Oreal Sustainability Challenge 2025 and TATA Crucible Campus Quiz 2025; Seed Hackathon Certificate recipient.
- Led AI literacy sessions under Student Social Responsibility (SSR) program, teaching prompt engineering and responsible AI usage to school students.
- Served as Crowd Management Coordinator at Amrita Varsham 2024, managing logistics for a large-scale campus event.

LANGUAGES

- Languages: Telugu (Native) | English (Professional Proficiency) | Hindi (Proficient) | Malayalam (Basic)